Unsupervised Natural Language Processing

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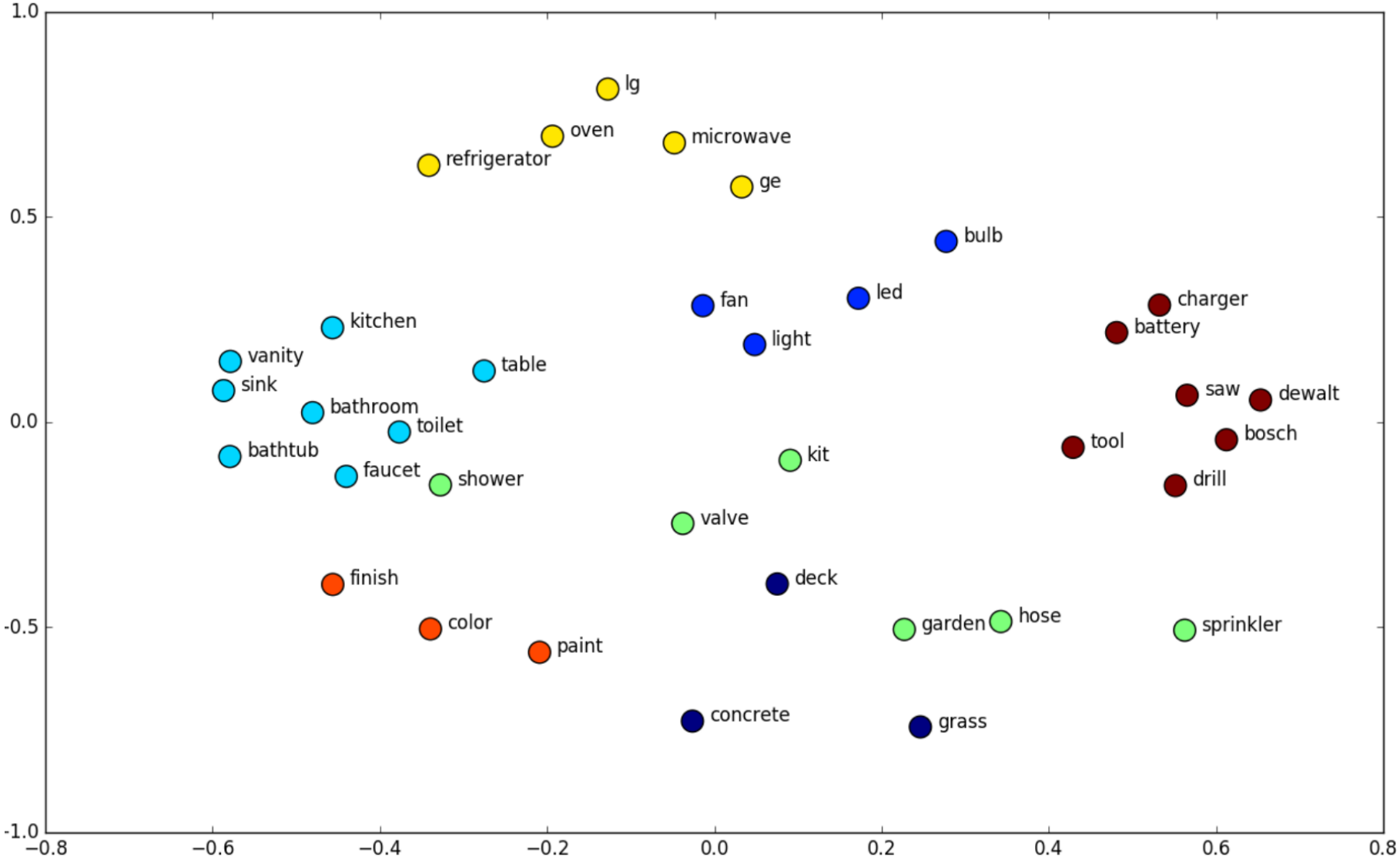
How to find clusters of documents?

- Vectorization + "classical" clustering
- TF-IDF
- LSA
- LDA

Warning : Topic modeling is very sensitive to pretreatments!

2D Visualization of high-dimensional words

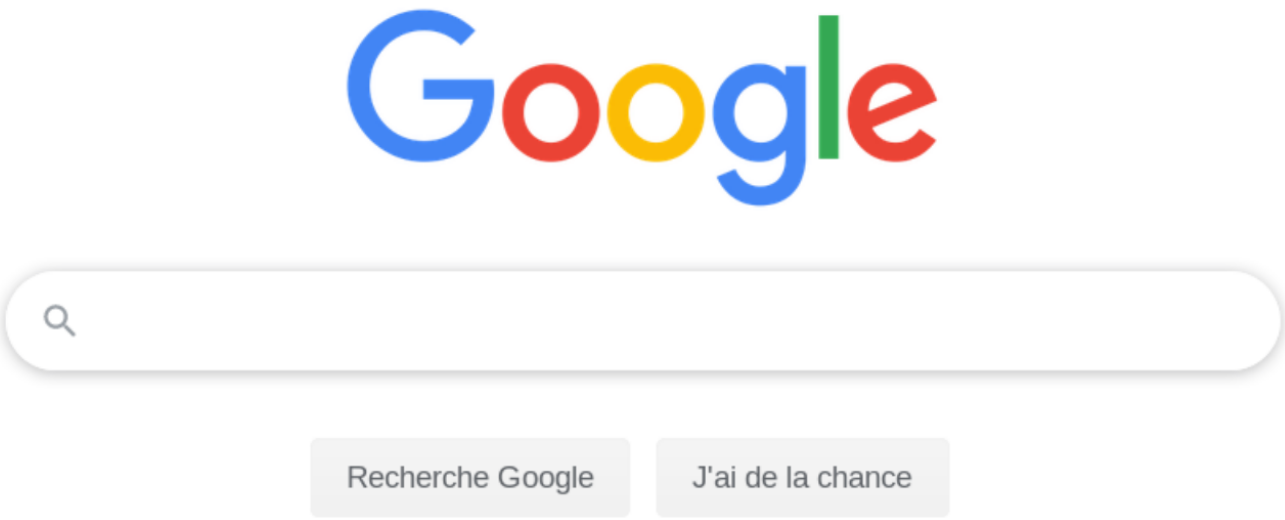


Word embedding
+
UMAP

Information Retrieval(IR)

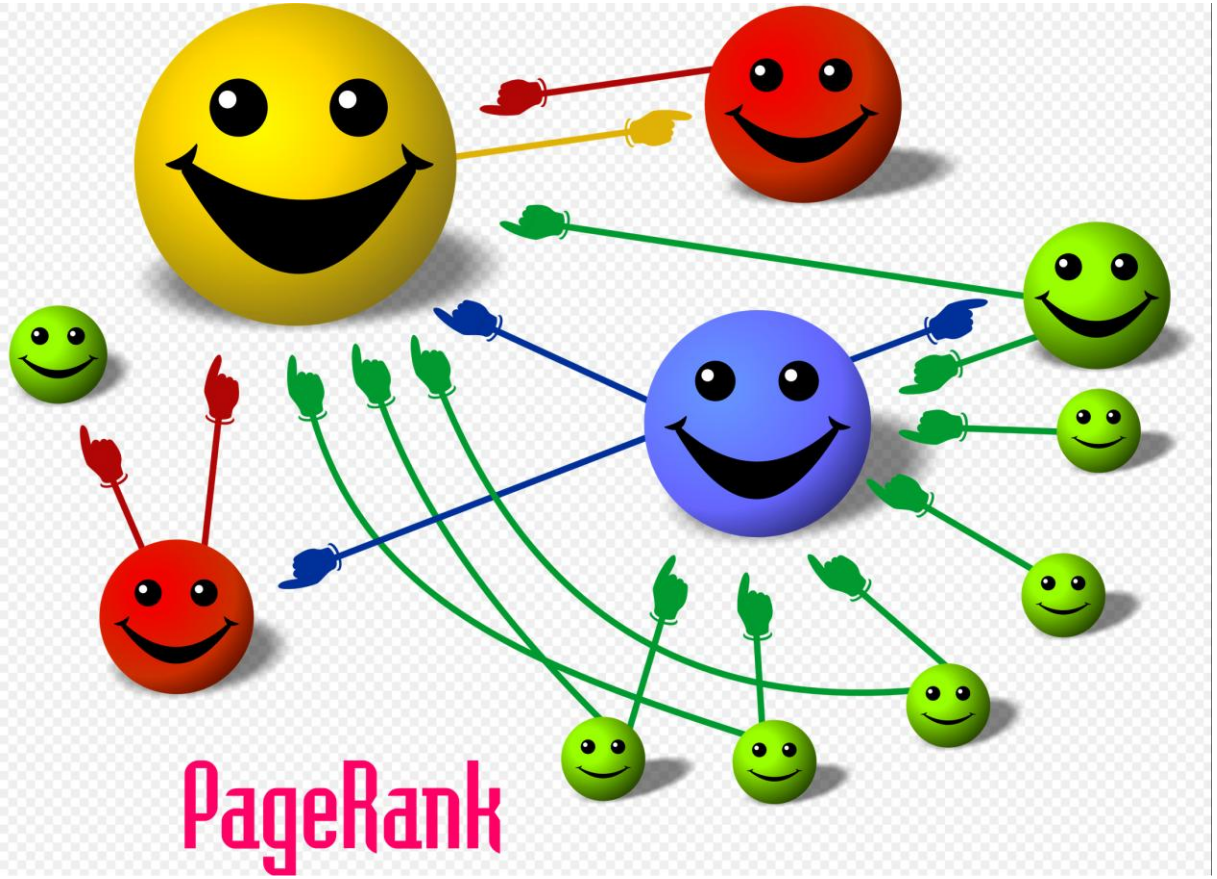
- IR is the process of retrieving the most suitable information as a response to the query being made by the user.

- Example:



The Anatomy of a Large-Scale Hypertextual Web Search Engine

Sergey Brin and Lawrence Page



Paper published in 98

IR Evaluation

- The accuracy of an information retrieval task is measured in terms of precision and recall.
- Suppose that a given IR system returns X documents when a query is fired. But the actual or gold set of documents that needs to be returned is Y.

IR Evaluation

- Recall may be defined as the fraction of gold documents that a system finds. It may be defined as the ratio of true positives to the combination of true positives and false negatives.

$$Recall\ (R) = (X \cap Y) / Y$$

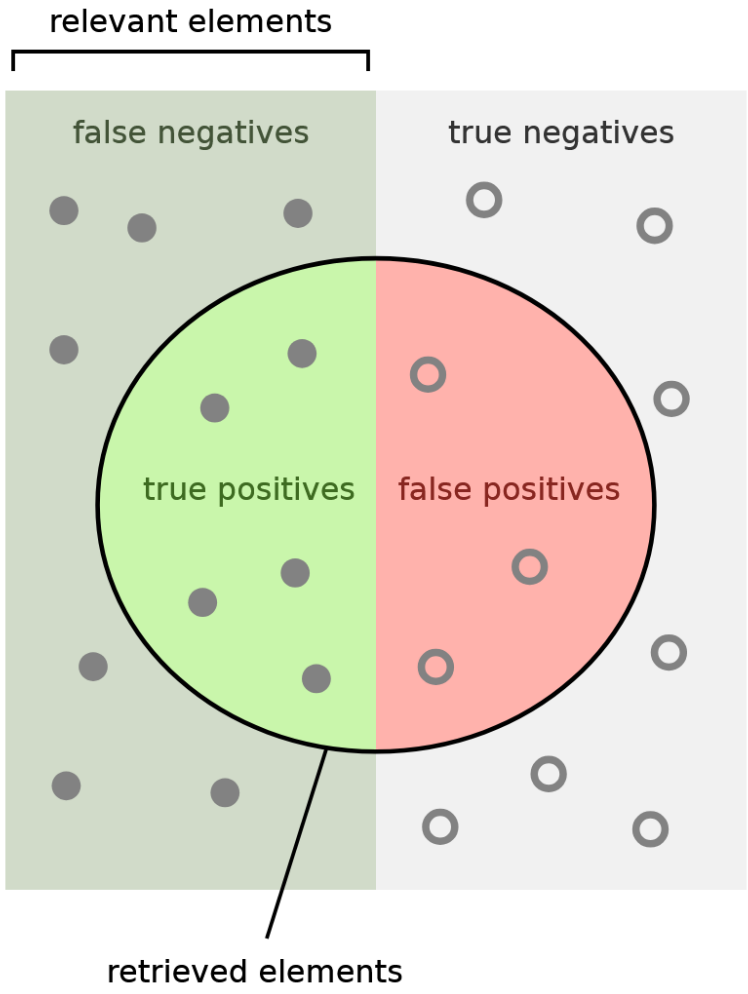
- Precision may be defined as the fraction of documents that an IR system detects and are correct.

$$Precision\ (P) = (X \cap Y) / X$$

- F-Measure may be defined as the harmonic mean of precision and recall.

$$F\text{-}Measure = 2 * (X \cap Y) / (X + Y)$$

IR Evaluation



How many retrieved items are relevant?

Precision =

How many relevant items are retrieved?

Recall =

TF-IDF

- **TF-IDF** = term frequency–inverse document frequency
- **TF or term frequency** : may be defined as the total number of times a given token exists in a document divided by the total number of tokens.

$$\mathrm{tf}(t,d) = 0.5 + 0.5 \cdot \frac{f_{t,d}}{\max\{f_{t',d} : t' \in d\}}$$

TF-IDF

- **IDF or inverse document frequency :** the specificity of a term can be quantified as an inverse function of the number of documents in which it occurs.

$$\mathrm{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|}$$

with

- N: total number of documents in the corpus $N = |D|$
- $|\{d \in D : t \in d\}|$: number of documents where the term t appears (i.e., $\mathrm{tf}(t, d) \neq 0$). If the term is not in the corpus, this will lead to a division-by-zero. It is therefore common to adjust the denominator to $1 + |\{d \in D : t \in d\}|$.

TF-IDF

$$\mathrm{tfidf}(t, d, D) = \mathrm{tf}(t, d) \cdot \mathrm{idf}(t, D)$$

Let’s see an example : https://en.wikipedia.org/wiki/Tf–idf

Beir

Benchmarking IR

- Link to the 2021 paper and 2024 paper
- Link to the GitHub

Some Limits of evaluation in NLP

- NLP is difficult because it's difficult to evaluate :

  "Beyond Accuracy: Behavioral Testing of NLP Models with CheckList" Best paper at ACL 2020

- Data Contamination

  " Investigating Data Contamination in Modern Benchmarks for Large Language Models" paper at ACL 2024

Data Contamination

Benchmark	ChatGPT		GPT-4		LLaMa 2-13B		Mistral-7B	
	EM	Rouge-L	EM	Rouge-L	EM	Rouge-L	EM	Rouge-L
PIQA (Bisk et al., 2020)	0.00	0.18	0.00	0.17	0.00	0.06	0.00	0.15
HellaSwag (Zellers et al., 2019)	0.00	0.13	0.02	0.12	0.00	0.04	0.00	0.09
OpenbookQA (Mihaylov et al., 2018)	0.01	0.13	0.01	0.13	0.04	0.08	0.10	0.19
WinoGrande (Sakaguchi et al., 2020)	0.09	0.10	0.12	0.13	0.01	0.01	0.03	0.01
TruthfulQA (Lin et al., 2022)	0.12	0.46	0.10	0.43	0.02	0.14	**0.15**	**0.61**
MMLU (Hendrycks et al., 2021)	**0.52**	0.69	**0.57**	0.67	0.00	0.06	0.01	0.12

Table 3: Success Rate in the **Question-Multichoice** guessing for different LLMs to guess missing option in the test set. Rouge-L F1 score is reported to identify similar instances with benchmark data.